

**Internet of Things
(SET/CS/BT/M802)
Program: B.Tech (CSE) – 8th Semester
Submission Mode: Written
Assignment 2**

Section A: Short Answer Questions

1. What is the role of the PHY layer in IoT communication?
2. Define MAC layer with one function.
3. What is IEEE 802.11 commonly used for?
4. Write one feature of IEEE 802.15.
5. What is Bluetooth Low Energy used for?
6. Define Zigbee.
7. What is the purpose of IPv4?
8. What is IPv6 and why is it needed?
9. Expand 6LoWPAN.
10. What is the role of RPL?
11. Define TCP and UDP with their functions.

Section B: Medium Answer Questions

1. Explain the functions of PHY and MAC layers in IoT systems.
2. Compare IEEE 802.11 and IEEE 802.15.
3. Describe the working of WirelessHART.
4. Explain the features of Z-Wave.
5. Differentiate between Bluetooth Low Energy and Zigbee.
6. Explain the concept of 6TiSCH.
7. Discuss the working of DHCP.
8. Explain the function of ICMP.
9. Explain TLS and DTLS.
10. Compare HTTP and CoAP.
11. Discuss the architecture of the Message Queuing Telemetry Transport Layer (MQTT) protocol with its function in IoT.

Section C: Long Answer Questions

1. Explain the classification of sensors with the help of a suitable diagram.
2. Discuss various IoT communication protocols such as Zigbee, Bluetooth Low Energy, WirelessHART, and Z-Wave with their applications.
3. Explain the network layer protocols in IoT including IPv4, IPv6, 6LoWPAN, and RPL.
4. Describe routing protocols in IoT such as RPL, CORPL, and CARP with suitable diagrams.
5. Explain session layer protocols in IoT such as MQTT, AMQP, XMPP, and CoAP.
6. Compare and contrast HTTP and MQTT in IoT applications.
7. Discuss the importance of security in IoT communication using TLS and DTLS.
8. Discuss any three application layer protocols used in IoT communication. Write the merits of using a modified OSI stack in an IoT and M2M system.

Bonus (Case Study / Application-Based)

1. Design an IoT-based smart home system and justify the use of Zigbee or Bluetooth Low Energy.
2. Suggest suitable protocols for a smart agriculture system and justify your choices.

Note: Students must complete and submit their assignments on time as per the given deadlines. Deadline for Assignment 2: 20/04/2026.
This is to inform all students that those who have not yet submitted Assignment No. 1 are no longer required to do so. However, students are strictly advised that failure to submit any future assignments will result in zero assignment marks. Please ensure the timely submission of all upcoming assignments. Instructions apply only to this subject (SET/CS/BT/M802).